LESSON 14 · CHEAT SHEET

The Charter Report

THE QUEEN'S QUESTIONS

Success rate	`outcome == "Success"` / all; Fled is **not** success
Difficulty	success % and injuries per quest, per 1–5
Worth it?	`sum(paid) / sum(days)` per monster
Regions	`LEFT JOIN regions`, warden or `'(none)'`
Loot	`party JOIN heroes`, sum of `loot_share`
Fair?	avg difficulty per class; leaders' share

PROJECT SHAPE

```
my_work/charter.py      # functions only, no print
my_work/check1.py       # import charter; print
my_work/report.py       # runs it all, writes files
python my_work/check1.py   # from lesson folder
```

`import charter` works because the script and the module sit in the same folder.

SQL INTO PANDAS

```
import sqlite3
con = sqlite3.connect("data/guild.db")
sql = """
    SELECT outcome, COUNT(*) AS quests
    FROM quests
    GROUP BY outcome
    ORDER BY quests DESC
"""
df = pd.read_sql(sql, con)
pd.read_sql("... LIMIT ?", con, params=(5,))
```

LEFT JOIN KEEPS THE LONELY ROWS

```
SELECT q.region,
       COALESCE(r.warden, '(none)') AS warden,
       COUNT(*) AS quests,
       SUM(q.outcome = 'Success') AS wins
FROM quests AS q
LEFT JOIN regions AS r
       ON q.region = r.region
GROUP BY q.region
```

`COUNT(*)` counts rows; `COUNT(r.warden)` skips NULLs. `1.0 *` avoids whole-number division.

A FUNCTION WITH A DOCSTRING

```
def success_rate(quests):
    """Share of Success in %, 1 decimal."""
    rate = quests["is_success"].mean() * 100
    return float(round(rate, 1))
```

PANDAS: PER GROUP

```
t = quests.groupby("monster").agg(
        quests=("quest_id", "count"),
        total_paid=("paid", "sum"),
        days=("days", "sum"))
t["paid_per_day"] = (t["total_paid"]
                     / t["days"]).round(1)
t = t.sort_values("paid_per_day",
                  ascending=False)
t.index[0]                  # best
t["avg_reward"].idxmax()    # label of max
```

MERGING AND PER-GROUP COUNTS

```
trips = (party.merge(heroes, on="hero_id")
              .merge(quests, on="quest_id"))
size = party.groupby("quest_id")["hero_id"] \
            .transform("count")   # per row
shared = party[size > 1]
```

OUTPUT (LESSON 13, CONDENSED)

```
cols = ["#c8553d" if m == "Wyrm"
        else "#1f3a5f" for m in s.index]
plt.barh(s.index, s.values, color=cols)
plt.savefig(OUT / "crowns_per_day.png")

with pd.ExcelWriter(path) as writer:
    summary.to_excel(writer, sheet_name="Summary",
                     index=False)
for ws in wb.worksheets:
    for cell in ws[1]:
        cell.font = Font(bold=True,
                         color="FFFFFF")
    ws.freeze_panes = "A2"
wb.create_sheet("Chart").add_image(
    Image(OUT / "crowns_per_day.png"), "B2")
```

GOTCHAS

- Fled is not Success. Count it, don't celebrate it.
- `reward` is a promise; `paid` is what came home.
- Plain `JOIN` drops Thornwood. `LEFT JOIN`.
- `round()` on a pandas number gives `np.float64`; wrap in `float()`.
- Changed `charter.py`? Re-run the check script; nothing reloads by itself.